From Geometry to a Real-Time Digital Twin of Rolling-Disc Notched Tendon-Driven Continuum Robots

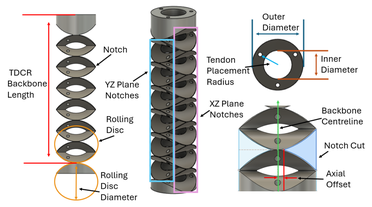
Om Mohan Pawar Supervisor: **Prof. Madhu Vadali** IITGN Robotics Laboratory, Mechanical Engineering, IIT Gandhinagar B.Tech Project, Final Evaluation, November 2026

IITGN ROBOTICS

PART A How does geometry govern compliance?

under review, ICRA 2027 — ICRA IEEE INTERNATIONAL CONFERENCE ON ROBOTICS AND AUTOMATION

The robot, and two questions



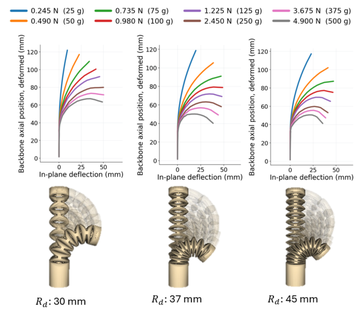
A. Design
Which geometry gives a wanted bend, and what stroke and force does it cost?

B. Model
Can the whole shape be predicted live from servo commands alone, for any stack and mounting?

TPU tube, two orthogonal families of rolling-disc notches, tendons in the wall, stackable 160 mm modules.

A1 SWEEP: geometry to simulation, unattended

Parameters D_{RD}, L, D_o, D_i, e, c, N_t → **CAD** CadQuery, one Boolean cut → **Mesh** CGAL tetrahedra → **FEM** SOFA, pull-only tendons → **Metrics** curvature, CC error, stroke



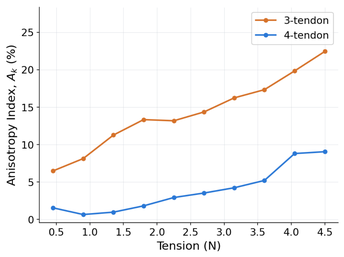
616
FEM runs: 14 designs × 4 tendons × 11 loads

56 MPa
measured TPU modulus, gravity on

$T \leq 0.9$ **N**
validity range of every claim

Same load, three disc diameters: a larger disc gives more, shallower notches and a tighter bend.

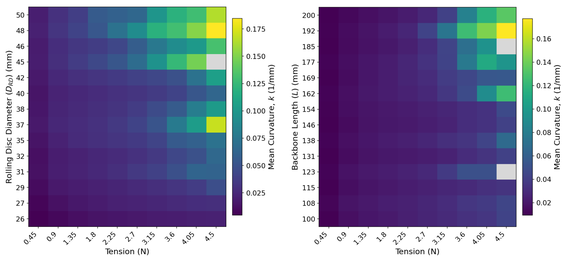
A2 Match the actuation to the structure



6.2 → 2.6 %
curvature spread across tendons $A_\kappa = \sigma_\kappa / |\bar{\kappa}|$, 3 → 4 tendons

2 motors
per section: four tendons form two antagonistic pairs

A3 What each parameter controls



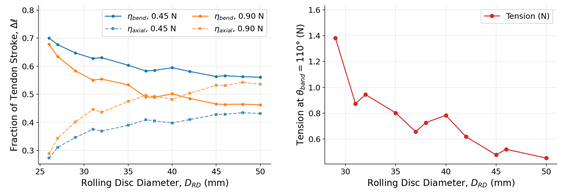
D_{RD} sets curvature
turning 54° → 143° (26 to 50 mm, 0.9 N)

L sets reach
tip travel 78 → 157 mm (100 to 200 mm)

Mean curvature (1/mm). More notches bend further, but deviate up to 5× more from constant curvature.

A4 Stroke budget: what a bend costs the actuator

$$\underbrace{\Delta\ell}_{\text{tendon stroke}} = \underbrace{\Theta\, d}_{\text{bending}} + \underbrace{\Delta L}_{\text{axial compression}}$$

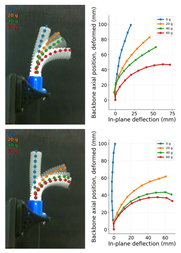


0.99
budget closure on notched designs

42 to 47 %
of the stroke is compression, not bending

0.87 → 0.45 N
for a 110° bend as D_{RD} grows

A5 Experiment: three printed prototypes



D_{RD} (mm)	NRMSE, % of length centreline	tip	axial
30	18.5	20.9	43.1
67.89	10.1	23.5	11.9
86.84	16.3	64.7	23.9

Shape agrees within 18.5 %; the tip less well: friction and self-contact are not modelled.

PART B A sensor-free, real-time digital twin

in preparation for IEEE RA-L — IEEE Robotics & Automation Society

B1 Servo commands in, whole shape out

commands $\delta\ell$ (servos) + mounting (gravity) → **Tendon map** $\delta\ell = G\lambda$ from CAD → **Pull-only** convex QP (slack) → **Gravity** fixed point, any angle → **Shape** pose every 5 mm

No sensor on the robot. One module model, composed by statics for any stack and mounting.

B2 The physics: a beam that follows the CAD

1 a tendon loads every slice — moment $M = Td$; compression $N = T$

2 it shortens and bends — $\varepsilon = aN$ $\kappa = CM$; tendon path shortens by $\varepsilon + d\kappa$

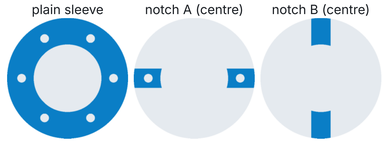
3 tendons only pull — slack $\lambda = 0$; taut $\lambda > 0$

$$\delta\ell = G\lambda, \quad G_{ij} = \sum_c \tau_{ic}\tau_{jc}\left(a_c + \boldsymbol{d}_i^{\top} C_c\, \boldsymbol{d}_j\right) + c_t L_i\, \delta_{ij}$$

$$\lambda^\star = \arg\min_{\lambda \geq 0} \tfrac{1}{2}\lambda^{\top} G\lambda - \delta\ell^{\top}\lambda$$

Tendon map: exactly linear, any bend

Pull-only: convex, unique, exact



÷ 8
area in a notch

÷ 280
bending stiffness: the notches do the bending

$\tau_{ic} \in \{0, 1\}$: tendon i passes slice c. $a_c = \int_c dy/A$, $C_c = \int_c dy/J$: slice compliances from the CAD. $c_t L_i$: tendon stretch.

B3 Identify the module law once, then verify

×20.2
axial notch compliance vs beam theory

×5.9 / 5.3
bending, two directions

×2.45
torsion (FEM twist test)

< 1 µm
solver vs closed-form beam cases

0.004 mm
vs an independent Cosserat rod

132 tests
one-command reproduction of every number

Law identified from 675 FEM poses (SOFA) of 1 and 2 modules, built from the same CAD.

B4 Scientific ML: learn the law, not the solution

Learned law θ $C_\theta(y) = C_{CAD}(y)\, e^{\theta}$ 4 physical numbers → **Exact physics** pull-only QP + gravity → **Shape** ⟷ loss → **Data** FEM now, mocap next

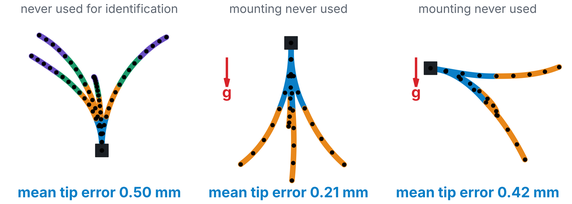
fit θ through the solver

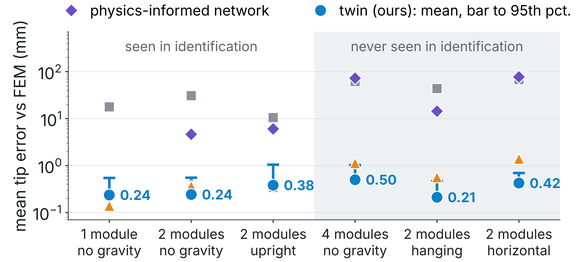
Next: a convex notch energy $\psi_\theta(N, \boldsymbol{M})$, with θ a function of geometry (Direction, step 2)

tip error vs FEM (mm)	seen	new mounting	4 modules
learn the **solution** (PINNs)	3 to 6	9 to 77	54 to 73, retrained
learn the law, in the solver	**0.2 to 0.4**	**0.2 to 0.4**	**0.50, same θ**

One θ for every stack and mounting. PINNs: forward and inverse, same data.

B5 Calibrate small, deploy large (FEM ground truth)





60 to 100 held-out FEM poses per set. Markers: mean; bar: 95th percentile.

0.50 mm
4 unseen modules, 0.08 % of length

1 to 4 ms
per pose (250 to 970 Hz)

62 → 0.5
mm: beam theory alone vs identified

B6 What the data can identify, and the fix

weight ↔ stiffness confounded → weigh modules, do not fit
equal pulls: distal tendons slack → pretension 1 : 1.75 : 2.25 : 2.5
0.4 mm moment-arm error → 5th number, recovered to 0.02 mm

Conclusions

- **Design:** disc diameter sets curvature, length sets reach; nearly half the stroke is compression.
- **Model:** a module law learned inside exact physics transfers to unseen stacks and mountings.
- **Limit:** Part B is validated against FEM only; hardware is next.

Direction A geometry-conditioned twin for inverse design

1 Hardware validation (now)
Motion capture, 4-module robot. Calibrate **5 numbers on 15 poses** of 2 modules; test 4 modules, upright and hanging.
Pass: tip error ≤ 2 % of length (13 mm), fixed before any data.

→

2 Law as a function of geometry
SWEEP generates designs; the twin identifies each: $\theta = f(D_{RD}, D_i, L)$.
Pass: beats CAD-only theory on designs held out entirely.

→

3 Inverse design
Target curvature + stroke budget → geometry, in milliseconds; print and verify.
If residuals demand: convex notch energy ψ_θ, friction, self-contact.

Outputs and my role

[1] D. Yadav, **O. M. Pawar**, Krishnapriya R, S. Gandhi, M. Vadali, "How Geometry Governs Compliance? A Parametric Study of Rolling-Disc Notched Tendon-Driven Continuum Robot," **under review, IEEE ICRA 2027**.

[2] O. M. Pawar et al., sensor-free digital twin of a modular notched TDCR, in preparation for IEEE RA-L, with an installable open package.

Role. Part A: SOFA 25.06 port, parallel sweeps, mesh and settling checks. Part B: model, solver, identification, calibration, baselines and software.

Builds on: Camarillo et al., T-RO 2008 (tendon mechanics) • Rucker & Webster, T-RO 2011 • Faure et al., SOFA, 2012 • Coevoet et al., SoftRobots, 2017 • Kim et al., arXiv:2507.10121 • Danesh et al., arXiv:2608.26273.

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